Free Response Solution Keys (By Part)

1 Problem 22

1.1 22(a)

1.1.1 Version A

22(a) Graphing Supply and Demand Curves

Invert the demand and supply equations (that is, write in "$P =$" form) and then graph the lines. It is clear from the "$P =$" form what the price intercept is.

Demand (inverse demand):

$$Q_D = 24 - 2P \implies 2P = 24 - Q \implies \boxed{P = 12 - \tfrac{1}{2}Q}.$$

So the demand price intercept is $\boxed{12}$.

Supply (inverse supply):

$$Q_S = 12 + 2P \implies 2P = Q - 12 \implies \boxed{P = -6 + \tfrac{1}{2}Q}.$$

So the supply price intercept is $\boxed{-6}$.

1.1.2 Version B

22(a) Graphing Supply and Demand Curves

Invert the demand and supply equations (that is, write in "$P =$" form) and then graph the lines. It is clear from the "$P =$" form what the price intercept is.

Demand (inverse demand):

$$Q_D = 26 - P \implies P = 26 - Q.$$

So the demand price intercept is $\boxed{26}$.

Supply (inverse supply):

$$Q_S = 10 + 3P \implies 3P = Q - 10 \implies P = -\tfrac{10}{3} + \tfrac{1}{3}Q.$$

So the supply price intercept is $\boxed{-\tfrac{10}{3}}$.

1.1.3 Version C

22(a) Graphing Supply and Demand Curves

Invert the demand and supply equations (that is, write in "$P =$" form) and then graph the lines. It is clear from the "$P =$" form what the price intercept is.

Demand (inverse demand):

$$Q_D = 30 - 2P \;\Rightarrow\; 2P = 30 - Q \;\Rightarrow\; \boxed{P = 15 - \tfrac{1}{2}Q}.$$

So the demand price intercept is $\boxed{15}$.

Supply (inverse supply):

$$Q_S = 18 + 2P \;\Rightarrow\; 2P = Q - 18 \;\Rightarrow\; \boxed{P = -9 + \tfrac{1}{2}Q}.$$

So the supply price intercept is $\boxed{-9}$.

1.1.4 Version D

22(a) Graphing Supply and Demand Curves

Invert the demand and supply equations (that is, write in "$P =$" form) and then graph the lines. It is clear from the "$P =$" form what the price intercept is.

Demand (inverse demand):

$$Q_D = 19 - P \;\Rightarrow\; P = 19 - Q.$$

So the demand price intercept is $\boxed{19}$.

Supply (inverse supply):

$$Q_S = 7 + 2P \;\Rightarrow\; 2P = Q - 7 \;\Rightarrow\; P = -\tfrac{7}{2} + \tfrac{1}{2}Q.$$

So the supply price intercept is $\boxed{-\tfrac{7}{2}}$.

1.1.5 Version E

22(a) Graphing Supply and Demand Curves

Invert the demand and supply equations (that is, write in "$P =$" form) and then graph the lines. It is clear from the "$P =$" form what the price intercept is.

Demand (inverse demand):

$$Q_D = 22 - 2P \;\Rightarrow\; 2P = 22 - Q \;\Rightarrow\; \boxed{P = 11 - \tfrac{1}{2}Q}.$$

So the demand price intercept is $\boxed{11}$.

Supply (inverse supply):

$$Q_S = 13 + P \;\Rightarrow\; P = Q - 13.$$

So the supply price intercept is $\boxed{-13}$.

1.2 22(b)

1.2.1 Version A

22(b) Equilibrium

Set the demand equation equal to the supply equation, and solve for P^*:

$$24 - 2P = 12 + 2P \implies 12 = 4P \implies \boxed{P^* = 3}.$$

Then $Q^* = 24 - 2(3) = \boxed{18}$.

1.2.2 Version B

22(b) Equilibrium

Set the demand equation equal to the supply equation, and solve for P^*:

$$26 - P = 10 + 3P \implies 16 = 4P \implies \boxed{P^* = 4}.$$

Then $Q^* = 26 - 4 = \boxed{22}$.

1.2.3 Version C

22(b) Equilibrium

Set the demand equation equal to the supply equation, and solve for P^*:

$$30 - 2P = 18 + 2P \implies 12 = 4P \implies \boxed{P^* = 3}.$$

Then $Q^* = 30 - 2(3) = \boxed{24}$.

1.2.4 Version D

22(b) Equilibrium

Set the demand equation equal to the supply equation, and solve for P^*:

$$19 - P = 7 + 2P \implies 12 = 3P \implies \boxed{P^* = 4}.$$

Then $Q^* = 19 - 4 = \boxed{15}$.

1.2.5 Version E

22(b) Equilibrium

Set the demand equation equal to the supply equation, and solve for P^*:

$$22 - 2P = 13 + P \implies 9 = 3P \implies \boxed{P^* = 3}.$$

Then $Q^* = 22 - 2(3) = \boxed{16}$.

1.3 22(c)

1.3.1 Version A

22(c) Q_t with tax $t = 2$ on sellers

To analyze imposing a tax on sellers, shift the inverse supply curve up by the amount of the tax, then use that shifted supply curve to find Q_t.

With a per-unit tax $t = 2$ on sellers, the inverse supply curve shifts up by 2:

$$P = \left(-6 + \tfrac{1}{2}Q\right) + 2 = -4 + \tfrac{1}{2}Q.$$

Set equal to inverse demand from part (a):

$$12 - \tfrac{1}{2}Q = -4 + \tfrac{1}{2}Q \;\Rightarrow\; 16 = Q \;\Rightarrow\; \boxed{Q_t = 16}.$$

1.3.2 Version B

22(c) Q_t with tax $t = 4$ on sellers

To analyze imposing a tax on sellers, shift the inverse supply curve up by the amount of the tax, then use that shifted supply curve to find Q_t.

With a per-unit tax $t = 4$ on sellers, the inverse supply curve shifts up by 4:

$$P = \left(-\tfrac{10}{3} + \tfrac{1}{3}Q\right) + 4 = \tfrac{2}{3} + \tfrac{1}{3}Q.$$

Set equal to inverse demand from part (a):

$$26 - Q = \tfrac{2}{3} + \tfrac{1}{3}Q \;\Rightarrow\; 78 - 3Q = 2 + Q \;\Rightarrow\; 76 = 4Q \;\Rightarrow\; \boxed{Q_t = 19}.$$

1.3.3 Version C

22(c) Q_t with tax $t = 4$ on sellers

To analyze imposing a tax on sellers, shift the inverse supply curve up by the amount of the tax, then use that shifted supply curve to find Q_t.

With a per-unit tax $t = 4$ on sellers, the inverse supply curve shifts up by 4:

$$P = \left(-9 + \tfrac{1}{2}Q\right) + 4 = -5 + \tfrac{1}{2}Q.$$

Set equal to inverse demand from part (a):

$$15 - \tfrac{1}{2}Q = -5 + \tfrac{1}{2}Q \;\Rightarrow\; 20 = Q \;\Rightarrow\; \boxed{Q_t = 20}.$$

1.3.4 Version D

22(c) Q_t with tax $t = 3$ on sellers

To analyze imposing a tax on sellers, shift the inverse supply curve up by the amount of the tax, then use that shifted supply curve to find Q_t.

With a per-unit tax $t = 3$ on sellers, the inverse supply curve shifts up by 3:

$$P = \left(-\frac{7}{2} + \frac{1}{2}Q\right) + 3 = -\frac{1}{2} + \frac{1}{2}Q.$$

Set equal to inverse demand from part (a):

$$19 - Q = -\frac{1}{2} + \frac{1}{2}Q \implies 38 - 2Q = -1 + Q \implies 39 = 3Q \implies \boxed{Q_t = 13}.$$

1.3.5 Version E

22(c) Q_t with tax $t = 3$ on sellers

To analyze imposing a tax on sellers, shift the inverse supply curve up by the amount of the tax, then use that shifted supply curve to find Q_t.

With a per-unit tax $t = 3$ on sellers, the inverse supply curve shifts up by 3:

$$P = (Q - 13) + 3 = Q - 10.$$

Set equal to inverse demand from part (a):

$$11 - \frac{1}{2}Q = Q - 10 \implies 21 = \frac{3}{2}Q \implies \boxed{Q_t = 14}.$$

1.4 22(d)

1.4.1 Version A

22(d) Buyers' price

To find the buyers' price, plug Q_t into the original demand function to find the P_b at which they demand Q_t.

Using $Q_D = 24 - 2P_b$ and $Q_t = 16$:

$$16 = 24 - 2P_b \implies 2P_b = 8 \implies \boxed{P_b = 4}.$$

1.4.2 Version B

22(d) Buyers' price

To find the buyers' price, plug Q_t into the original demand function to find the P_b at which they demand Q_t.

Using $Q_D = 26 - P_b$ and $Q_t = 19$:

$$19 = 26 - P_b \implies \boxed{P_b = 7}.$$

1.4.3 Version C

22(d) Buyers' price

To find the buyers' price, plug Q_t into the original demand function to find the P_b at which they demand Q_t.

Using $Q_D = 30 - 2P_b$ and $Q_t = 20$:

$$20 = 30 - 2P_b \implies 2P_b = 10 \implies \boxed{P_b = 5}.$$

1.4.4 Version D

22(d) Buyers' price

To find the buyers' price, plug Q_t into the original demand function to find the P_b at which they demand Q_t.

Using $Q_D = 19 - P_b$ and $Q_t = 13$:

$$13 = 19 - P_b \implies \boxed{P_b = 6}.$$

1.4.5 Version E

22(d) Buyers' price

To find the buyers' price, plug Q_t into the original demand function to find the P_b at which they demand Q_t.

Using $Q_D = 22 - 2P_b$ and $Q_t = 14$:

$$14 = 22 - 2P_b \implies 2P_b = 8 \implies \boxed{P_b = 4}.$$

1.5 22(e)

1.5.1 Version A

22(e) Incidence of tax

To determine incidence, compare the change in buyers' price and sellers' price relative to the pre-tax equilibrium.

Sellers receive $P_s = P_b - t = 4 - 2 = \boxed{2}$.

Before the tax, $P^* = 3$. After the tax, buyers pay 4 (up $+1$) and sellers receive 2 (down -1). Thus: **Tax burden split evenly by buyers and sellers.**

1.5.2 Version B

22(e) Incidence of tax

To determine incidence, compare the change in buyers' price and sellers' price relative to the pre-tax equilibrium.

Sellers receive $P_s = P_b - t = 7 - 4 = \boxed{3}$.

Before the tax, $P^* = 4$. After the tax, buyers pay 7 (up $+3$) and sellers receive 3 (down -1). Thus: **Tax burden falls more heavily on buyers.**

1.5.3 Version C

22(e) Incidence of tax

To determine incidence, compare the change in buyers' price and sellers' price relative to the pre-tax equilibrium.

Sellers receive $P_s = P_b - t = 5 - 4 = \boxed{1}$.

Before the tax, $P^* = 3$. After the tax, buyers pay 5 (up $+2$) and sellers receive 1 (down -2). Thus: **Tax burden split evenly by buyers and sellers.**

1.5.4 Version D

22(e) Incidence of tax

To determine incidence, compare the change in buyers' price and sellers' price relative to the pre-tax equilibrium.

Sellers receive $P_s = P_b - t = 6 - 3 = \boxed{3}$.

Before the tax, $P^* = 4$. After the tax, buyers pay 6 (up $+2$) and sellers receive 3 (down -1). Thus: **Tax burden falls more heavily on buyers.**

1.5.5 Version E

22(e) Incidence of tax

To determine incidence, compare the change in buyers' price and sellers' price relative to the pre-tax equilibrium.

Sellers receive $P_s = P_b - t = 4 - 3 = \boxed{1}$.

Before the tax, $P^* = 3$. After the tax, buyers pay 4 (up $+1$) and sellers receive 1 (down -2). Thus: **Tax burden falls more heavily on sellers.**

2 Problem 23

2.1 23(a)

2.1.1 Version A

23(a) Output at $P = 4$

At price $P = 4$, each firm produces up to the quantity where marginal cost is weakly below price.

Firm A supplies $\boxed{4}$ units (since MC $1, 2, 3, 4 \leq 4$).

Firm B supplies $\boxed{2}$ units (since MC $2, 4 \leq 4$ and the next unit has MC $6 > 4$).

2.1.2 Version B

23(a) Output at $P = 3$

At price $P = 3$, each firm produces up to the quantity where marginal cost is weakly below price.

Thus $\boxed{Q_A = 3}$ and $\boxed{Q_B = 1}$.

2.1.3 Version C

23(a) Output at $P = 6$

At price $P = 6$, each firm produces up to the quantity where marginal cost is weakly below price.

Thus $\boxed{Q_A = 3}$ and $\boxed{Q_B = 2}$.

2.1.4 Version D

23(a) Output at $P = 8$

At price $P = 8$, each firm produces up to the quantity where marginal cost is weakly below price.

Thus $\boxed{Q_A = 4}$ and $\boxed{Q_B = 2}$.

2.1.5 Version E

23(a) Output at $P = 4$

At price $P = 4$, each firm produces up to the quantity where marginal cost is weakly below price.

Thus $\boxed{Q_A = 1}$ and $\boxed{Q_B = 4}$.

2.2 23(b)

2.2.1 Version A

23(b) Cost-minimizing allocation for total $Q = 6$

To minimize total cost, allocate production to the lowest marginal cost units across both firms.

The 6 lowest marginal costs are:
$A : 1, 2, 3, 4$ and $B : 2, 4$.

Thus the cost-minimizing allocation is $\boxed{Q_A = 4}$ and $\boxed{Q_B = 2}$. Note that at those quantities, $MC_A = MC_B$.

2.2.2 Version B

23(b) Cost-minimizing allocation for total $Q = 4$

To minimize total cost, allocate production to the lowest marginal cost units across both firms.

The 4 lowest marginal costs are $A : 1, 2, 3$ and $B : 3$.

Thus the cost-minimizing allocation is $\boxed{Q_A = 3}$ and $\boxed{Q_B = 1}$.

2.2.3 Version C

23(b) Cost-minimizing allocation for total $Q = 5$

To minimize total cost, allocate production to the lowest marginal cost units across both firms.

Thus the cost-minimizing allocation is $\boxed{Q_A = 3}$ and $\boxed{Q_B = 2}$.

2.2.4 Version D

23(b) Cost-minimizing allocation for total $Q = 6$

To minimize total cost, allocate production to the lowest marginal cost units across both firms.

Thus the cost-minimizing allocation is $\boxed{Q_A = 4}$ and $\boxed{Q_B = 2}$.

2.2.5 Version E

23(b) Cost-minimizing allocation for total $Q = 4$

To minimize total cost, allocate production to the lowest marginal cost units across both firms.

The 4 lowest marginal costs are $B : 1, 2, 3$ and then $A : 3$ (cheaper than B's next unit at MC 4).

Thus the cost-minimizing allocation is $\boxed{Q_A = 1}$ and $\boxed{Q_B = 3}$.